

PRODUCT SPECIFICATION SHEET

Product name	LUX-SP210 Portable Bluetooth Speaker Lamp	Supplier	Ningbo Lumina Acoustics
Product reference	LSP210-BLK	Factory	Ningbo, Zhejiang, CN
Customer	Meridian Home Brands LLC	PO No.	PO-88231
Date	22 January 2025	IP rights	Meridian Home Brands

This document is Meridian Home Brands LLC's property and defines the approved specification for the referenced product.

REVISION HISTORY

Rev.	Description of change	Page No.	Author	Date
0	Document creation	--	W. Zhang	05 Dec 2024
1	Added audio driver tuning results & cost breakdown	3	W. Zhang	18 Dec 2024
2	Retail carton artwork header confirmed	5	S. Whitman	22 Jan 2025

TABLE OF CONTENTS

- 1. Product
 - 1.1 General Specifications
 - 1.2 Dimensions, Weight & Materials
 - 1.3 On-Site Tests on Product
 - 1.4 Photos of Approved Sample
- 2. Inner Packing
 - 2.1 General Specifications
 - 2.2 Photos of Approved Inner Packing
- 3. Export Packing
 - 3.1 General Specifications
 - 3.2 Photos of Approved Export Packing
- 4. Defects Classification

1. PRODUCT

1.1 GENERAL SPECIFICATIONS

Checkpoint	Requirement	Sampling
Product function	Portable rechargeable table lamp with integrated Bluetooth loudspeaker. Dual-function: ambient LED lighting + wireless audio playback.	100%
Audio system	3 W full-range driver · Bluetooth 5.0 wireless · tuned aluminium enclosure for resonance control. Frequency response 120 Hz – 18 kHz; SNR ≥ 80 dB.	AQL 2.5
Wireless / connectivity	Bluetooth 5.0, profile A2DP/AVRCP; pairing range ≥ 10 m line-of-sight. Auto-reconnect to last device. Latency ≤ 200 ms.	AQL 2.5
Lighting system	Dimmable LED ring · 3 brightness settings (low / medium / high) · frosted polycarbonate diffuser for even, glare-free output. Colour temp 2700 K warm white; CRI ≥ 80.	AQL 2.5
Power system	5,000 mAh Li-ion · USB-C charging (5 V/2 A). Runtime ≥ 8 h audio at 50% vol; ≥ 12 h lighting at medium. Full charge ≤ 3.5 h. UN38.3 certified cell.	100%
Enclosure / finish	Anodised aluminium body (tuned as acoustic chamber), matte black. No sharp edges, burrs or flashing. Colour to approved Pantone swatch.	AQL 2.5
Controls & indicators	Top-mounted capacitive touch: power, volume ±, light toggle. 4-LED battery indicator. USB-C port on base with rubber gasket.	AQL 2.5
Cost contribution (BOM)	Audio 55% · power 25% · lighting 20% — reference split of unit component cost, used for classification & valuation support.	Ref. only
Markings / labelling	“Made in China”, model LUX-SP210, battery rating, CE/FCC/UKCA marks moulded or printed on base. Barcode label per customer spec.	100%
Compliance	CE, FCC Part 15B, UKCA, RoHS, REACH. Battery UN38.3 & UN3481 (PI 967). RF per EN 300 328.	Doc. review

1.2 DIMENSIONS, WEIGHT & MATERIALS

Checkpoint	Requirement	Sampling
Total product size	Ø 90 mm × 165 mm (H). Tolerance ± 2 mm.	AQL 2.5
Total product net weight	0.95 kg ± 30 g per unit.	AQL 2.5
Main body material	6063 anodised aluminium, 1.2 mm wall.	AQL 2.5
Diffuser material	Frosted polycarbonate (PC), UV-stabilised.	AQL 2.5
Base material	ABS with TPU anti-slip ring.	AQL 2.5
Driver	3 W full-range, 45 mm, neodymium magnet.	100%
Battery	Li-ion 18650 × 1, 5,000 mAh, 3.7 V nominal.	100%

1.3 ON-SITE TESTS ON PRODUCT

Test name	Test description & expected result	Sampling
Power-on / function	Power on; confirm LED ring illuminates and Bluetooth is discoverable. Expected: boots < 3 s, all functions respond.	100%
Bluetooth pairing	Pair with reference phone at 10 m. Expected: connects < 5 s, audio streams without dropout for 2 min.	AQL 2.5
Audio quality	Play reference track at 50%/80% volume. Expected: no rattle, distortion or enclosure buzz; balanced output.	AQL 2.5
Light dimming	Cycle all 3 brightness levels. Expected: distinct, even steps; diffuser shows no hotspots or dark bands.	AQL 2.5
USB-C charge	Insert USB-C cable; confirm charge indicator active. Expected: charging LED on, no port looseness.	AQL 2.5
Battery runtime (audit)	Sample units run to depletion. Expected: ≥ 8 h audio @ 50% volume.	1 pc/lot
Drop test	Drop from 1.0 m onto concrete, 1 face + 1 corner. Expected: no functional failure, no cracked diffuser.	1 pc/lot
Hi-pot / safety	Charging circuit insulation check. Expected: pass per IEC 60335 leakage limits.	1 pc/lot

1.4 PHOTOS OF APPROVED SAMPLE(S)

Description	Reference
Front / lit — LED ring on, diffuser glowing	<i>[approved sample photo — front, illuminated]</i>
Top — capacitive touch controls & battery LEDs	<i>[approved sample photo — top control surface]</i>
Base — USB-C port, rating label, barcode	<i>[approved sample photo — base & labelling]</i>

Reference sample	Identified with (label / tag / stamp)	Signed by
☒ Already in factory ☐ To be sent to inspectors ☐ No ref. sample	Gold seal sample #GS-LSP210-01, red QC tag, signed & dated.	W. Zhang (QC) / S. Whitman (Meridian)

2. INNER PACKING

2.1 GENERAL SPECIFICATIONS

Checkpoint	Requirement	Sampling
Retail box type	Individual printed colour box, corrugated E-flute, full-colour litho-laminated.	AQL 2.5
Retail box header	“Portable Bluetooth Speaker” — primary front-panel product name/header on retail carton artwork. Model LUX-SP210 shown beneath.	100%
Box artwork	Front: product hero image + header. Back: feature icons (3 W audio, BT 5.0, dimmable light, USB-C, 5,000 mAh). Multi-language.	AQL 2.5
Inner protection	Moulded pulp tray + PE foam top pad. Product in non-woven drawstring bag.	AQL 2.5
Accessories in box	1 × USB-C cable (1.2 m), 1 × user manual (multi-lang), 1 × warranty/safety card.	100%
Retail box dimensions	110 × 110 × 190 mm. Tolerance ± 3 mm.	AQL 2.5
Gross weight (boxed)	1.15 kg ± 40 g.	AQL 2.5
Box markings	Barcode (EAN-13), model, battery warning label (UN3481), country of origin.	100%

2.2 PHOTOS OF APPROVED INNER PACKING

Description	Reference
Retail box front — “Portable Bluetooth Speaker” header visible	[approved inner-packing photo — front panel]
Retail box back — feature icons & specs	[approved inner-packing photo — back panel]
Open box — pulp tray, product in bag, accessories	[approved inner-packing photo — contents]

3. EXPORT PACKING

3.1 GENERAL SPECIFICATIONS

Checkpoint	Requirement
Quantity & breakdown inside export carton	15 retail boxes per export carton. Speaker-lamp SKU packed 15/carton; total shipment 2,400 pcs = 160 export cartons.
Export carton dimensions	580 × 360 × 400 mm. Tolerance ± 5 mm.
Export carton weight	Net 14.25 kg · gross ≈ 15.6 kg per carton.
Export carton quality	5-ply corrugated (AB-flute), ≥ 250 g/m ² kraft liner. Edge crush ≥ 8 kN/m.
Inside protection	Retail boxes in 3×5 grid, corrugated dividers between layers; corner posts on top layer.
Carton sealing	H-taped top & bottom with 60 mm reinforced tape; 2 PP straps around girth.
Shipping marks	MERIDIAN / EDISON NJ, C/NO 1-160, MADE IN CHINA, gross/net wt, dimensions, handling & Li-ion (UN3481) labels.
Palletisation	1 × 20' FCL, floor-loaded (or 40 cartons/pallet × 4 if palletised). Total gross 3,120 kg / net 2,760 kg for full shipment.

3.2 PHOTOS OF APPROVED EXPORT PACKING

Description	Reference
Sealed export carton with shipping marks	[approved export-packing photo — sealed carton]
Open carton — retail boxes & dividers layout	[approved export-packing photo — internal layout]
Container / pallet loading	[approved export-packing photo — FCL loading]

4. DEFECTS CLASSIFICATION

Defects are classified against the approved sample. AQL levels: Critical 0 · Major 2.5 · Minor 4.0.

Description of defect	Critical	Major	Minor
Unit does not power on / no audio / no light	×		
Battery overheats, swells, or leaks	×		
USB-C charging non-functional	×		
Bluetooth will not pair or drops repeatedly		×	
Audible rattle, buzz or distortion from enclosure		×	
LED dimming steps missing or diffuser hotspot		×	
Runtime materially below 8 h audio spec		×	
Wrong or missing retail box header / artwork		×	
Missing accessory (cable / manual)		×	
Minor scratch or blemish on aluminium body			×
Slight colour deviation from approved swatch			×
Print registration off on retail box			×
Loose or crooked barcode label			×

----- End of document -----